

IMMUNOASSAY:

Immunoassay techniques are important for the specific detection of hormones, drugs, vitamin, and other compounds at nanogram and smaller levels. In this techniques, an antigen and a (specific) antibody react to form a complex or precipitate. Immunoassay techniques generally involve a competitive reaction b/w an analyte antigen and a standard antigen that has been tagged, for limited binding sites on the antibody to that antigen. The tag may be a radioactive tracer, an enzyme, or a fluorophore. Immunoassay can be very selective and sensitive.

Principle of Immunoassay

Immunoassay combines the sensitivity of radiochemistry, Fluorescence, or enzymatic tags with the specificity of immunology. Immunology is the study of antigens and their reactions with antibodies, that is, an organism's defense mechanism to foreign bodies through antibodies. An antigen (e.g. a hormone) is a foreign substance capable of producing antibody formation in the body and is able to react with (bind to) that antibody. An antibody is a protein endowed with the capacity to recognize a substance foreign to the organism it has invaded e.g. bacteria and viruses. When protein exhibit antibody activity, it is referred to as an immunoglobulin (Ig). The antibody will specifically react with an antigen to form antigen-antibody complex. An antibody is produced for use in immunoassay by injecting the antigen into an animal species to which it is foreign and recovering the serum that contains the resultant antibody (anti-serum).

The strength of the antigen-antibody complex is called the affinity or avidity. Affinity refers to the intrinsic association constant between an antibody and a univalent antigen, while avidity refers to the overall binding energy plus antibodies and multivalent antigen. We can write the overall binding reaction as $Ab + Ag \rightleftharpoons AbAg$

$$\text{formation Constant } K = \frac{[AbAg]}{[Ab][Ag]}$$

Principle: The analyte antigen competes with the tagged antigen for sites on the antibody. The displaced tagged antigen is measured, either directly or indirectly.

All immunoassay procedures are based on the original discovery by Berson and Yalow that low conc. of antibodies to the antigen hormone insulin could be detected radiochemically by their ability to bind radiolabeled (^{125}I) insulin. The determination of unknown conc. of antigen is based on the fact that radiolabeled antigen and unlabeled antigen (from the sample or standard) compete physiochemically for the binding sites on the antibodies (radioimmunoassay, RIA). The initial reaction vessel contains antibody solution (antiserum), labeled antigen, and the serum sample that may contain unlabeled (natural) antigen which is the substance to be determined. Upon incubation, the antibody (Ab) will form an antigen-antibody immune complex ($Ag-Ab$). In the absence of unlabeled antigen, a certain fraction of the labeled antigen (Ag^*) is bound as (Ag^*-Ab).

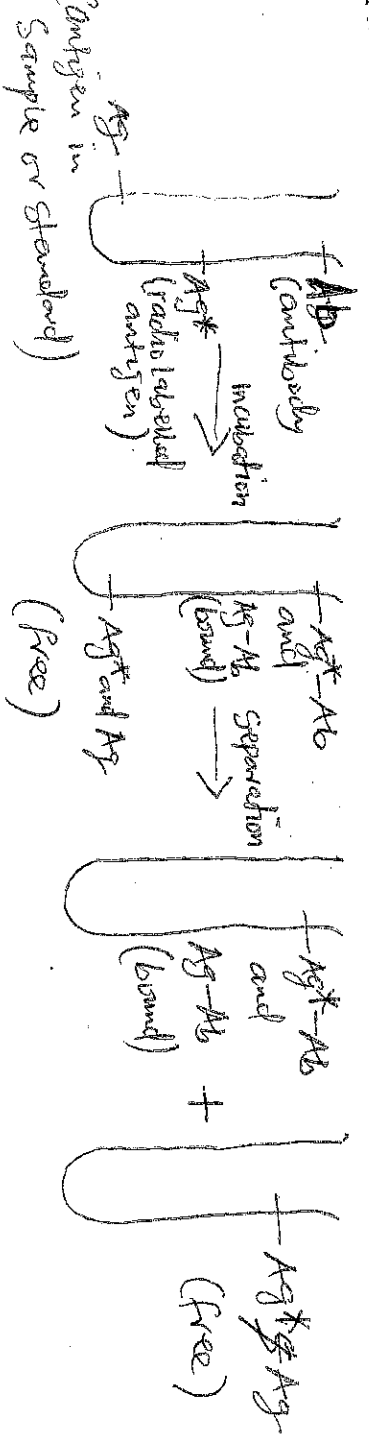


Figure 1. Principle of radioimmunoassay.

But when the increasing amount of unlabelled antigen (Ag) are added, the limited binding sites of the antibody are progressively saturated and the antibody can bind less of the radio-labelled antigen. After incubation, the bound antigen are separated from the unbound (free) antigen, and the labeled portion phases is measured to determine the percent bound of the labeled antigen. A diminished binding of labeled antigen when sample is added indicates the presence of unlabelled (natural antigen). A calibration curve is prepared using antigen standard of known conc. by plotting either the percent bound labeled antigen or the ratio of percent bound to free (B/F) as a function of the unlabelled antigen concentration. From this, the unknown antigen concentration can be ascertained.

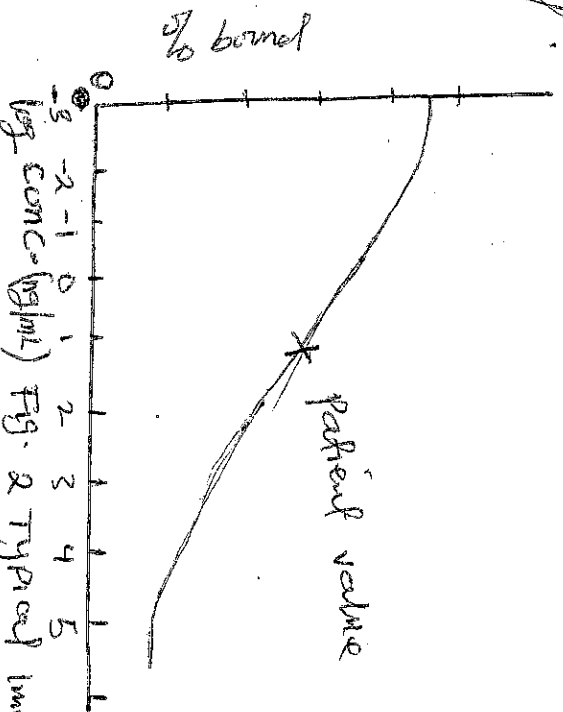


Fig. 2 Typical immunoassay calibration curve

Types of labeled used in immunoassay

(i) Radiolabeled: - This is a process where a radioactive substance is allowed to an antigen. ^{Antibodies} low conc. of antibodies to the antigen hormone ^{insulin} could be detected radiochemically by their ability to bind radiolabeled (^{125}I) insulin. The determination of unknown conc. of antigen is based on the fact that radiolabeled antigen and unlabeled antigen compete physiologically for binding sites on the antibodies this is called radioimmunoassay (RIA).

(ii) Fluorescence immunoassay labels: - Antibodies or antigen can be labeled with fluorescent dyes and the fluorescence used for measurement. Fluorescein isothiocyanate (FITC) and lissamine rhodamine B (RB200) are commonly used dyes. The protein to be conjugated with the dye is usually a mixture of immunoglobulin and should be free of other serum protein fraction like albumin and other globulins labeled more readily than gamma globulin and would render the method non specific. After labeling, unreacted dye is removed and the labeled protein purified using size exclusion chromatography.

(iii) Enzymes: -

Here, the ~~top~~ employed is an enzyme e.g. peroxidase, and the activity of the ^{unreacted} enzyme-tagged antigen is measured using a appropriate enzyme reaction. Homogenous enzyme immunoassay are commonly used for measurement of low-molecular weight molecules such as digoxin, amphetamine, and prescription drugs. Enzyme immunoassays are known as enzyme-linked immunosorbent assays (ELISA).

Separation of bound and free antigen

(i) PRECIPITATION:-

This is the widely used separation techniques to separate bound and free antigen. High conc. of various salts e.g. $(\text{NH}_4)_2\text{SO}_4$, Na_2SO_4 or organic solvents e.g. acetone, ethanol to precipitate the antigens. The precipitated complex is generally separated by centrifugation and filtration, and then the label response in either of the phases can be measured.

(ii) DOUBLE-ANTIBODY TECHNIQUE:-

Here, a second antibody is employed to precipitate the primary antigen-antibody complex. This second antibody is produced by injecting a second animal with gamma globulin produced in the first animal used to prepare the first antibody. Although the use of second animal introduces another source of error, the double-antibody technique has the advantage of being applicable for almost any immunoassay, and the separation is complete.

(iii) ELECTROPHORESIS:-

An electrical current is passed into the mixture of antigen and antibodies, this leads to the migration of charge species at different rates. The faster charge species are separated first followed by the slower species.

AFFINITY CHROMATOGRAPHY:-

(iv) Bonding of the antigen or antibody to a solid phase for use as reagent. They reversibly and selectively bind to analyte molecules. The mixture passes through the column, only the molecules that selectively bind to the affinity ligand are retained. Molecules that do not bind pass through the column with the mobile phase.

After the undesired molecules are removed, the retained antigen can be eluted by changing the mobile-phase conditions. This separation has the major advantage of extraordinary specificity.

Advantages of immunoassay

- It is very Sensitive
 - It is highly selective because of antigen and antibody selective binding can
 - Fluorescence immunoassay can measure homogeneous mixture of antigens
- ### Disadvantage
- The reaction takes time to reach equilibrium. Some may take hours.
 - Higher concentration of reagents may decrease the activity of enzymes.

PRINCIPLES OF AUTOMATION

There are two basic types of automation equipment, automatic devices and automated devices.

Automatic devices:- perform specific operations at a given point in an analysis, mostly the measurement step.

Thus, an automatic titrator will stop a titration at the end point, either mechanically or electrically, upon sensing a change in the property of the solution.

Automated devices:- control and regulate a process without human intervention. They do so through mechanical and electronic devices that are regulated by means of feedback information from a sensor.

Hence, an automated titrator may maintain a sample pH at a preset level by addition of acid or base as slight changes from the set pH is sensed by a pH electrode. Automated devices are widely used in process control systems, whereas automatic instruments are used in the analytical for performing analyses.

Discrete Analyzers

In discrete analyzers, a batch sample is taken at selected intervals and then analysed, with the information being fed to the controller and operator. Short transient errors may be detected during the measurement interval and a correction for this effected during the entire interval between measurement. Discrete measurement must be made when the sensing instrument requires discrete samples, as in chromatography, or a flow injection analyzer.

Continuous Flow Analyzers

Continuous ~~flow~~ - Process control instrument may make measurement directly in a flowing stream or a batch process reactor such as fermenter. Direct sensing devices such as electrode must be used. If a sample dilution, temperature control, or reagent addition is required, or measurement are with nonprobe-type instruments. Then, a small fraction of the stream is diverted into a feed stream where reagent may be mixed continuously and automatically with the sample, and the test measurement is made. Process control instrument operates by means of a control loop, which consists of sensor, controller, and operator.

Flow Injection Analysis (FIA).

In this process, samples are injected from a sample loop into a flowing stream containing one or more reagents. The sample plug is allowed to disperse in a controlled manner before it reaches the detector. Sample dispersion is controlled by the sample size, the flow rate, and the length and diameter of the tubing. The detector continuously records the absorbance, electrode potential, or other physical parameter as it changes as a result of the passage of the sample material through the flow cell.

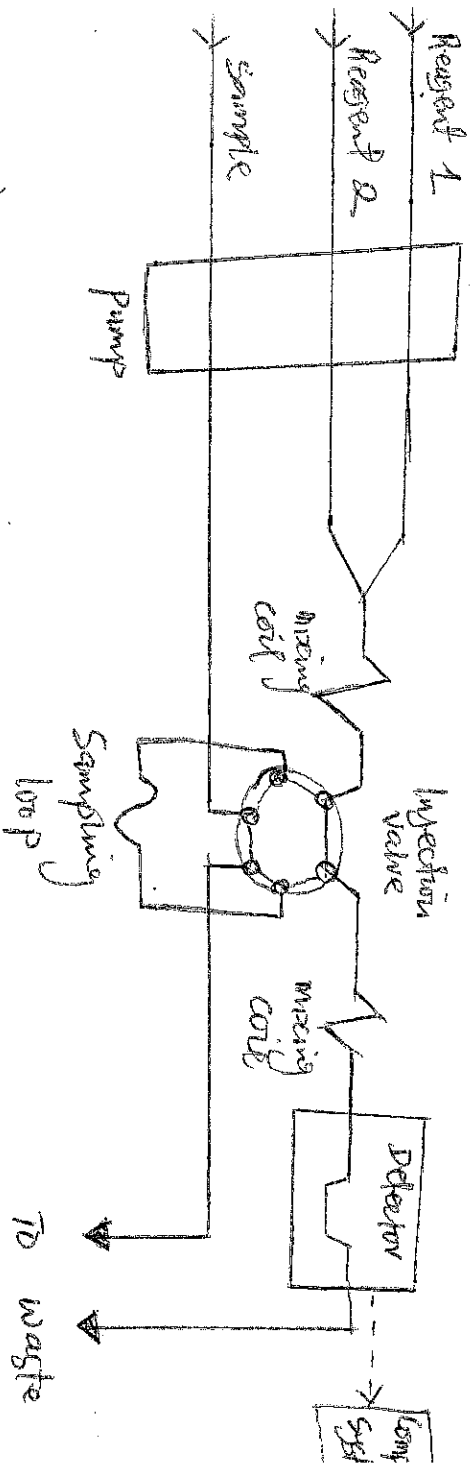


Figure 3. Flow Injection Analysis system.

Automated Sample Handling

Because of Once sampling has been accomplished and the no of samples and replicates chosen, sample processing begins. Because of their ~~cost~~ reliability and cost effectiveness, many laboratory are using automated sample handling methods. In some cases, automated sample handling is used only for a few specific operations, such as digesting the sample and removing interferences; in other cases

all the remaining steps in analytical procedure are automated. Automated sample handling can lead to ~~higher~~ more analyses per unit time, higher reliability, and lower cost than manual sample handling. Two different methods of automated sample handling are batch or discrete method and the continuous flow method.

(i) Batch or Discrete Method

System that process sample in a discrete manner resemble the operation that may be performed manually. Some discrete sample processors automate only the measurement steps of the procedures or a few chemical steps and measurement step. One type, base on the use of centrifugal force, mixes sample and reagents and transfer them to a photometric instrument for measurement. Another type, based on multilayer film technology, carries out a series of chemical reaction or physical processes in a sequential manner.

(ii) Continuous Flow Method

In continuous flow methods, the sample is injected into a flowing stream, where a number of operations can be performed before it is transported to a flow-through detector. These systems behave as automated analyzers in that they can perform not only sample processing operations but also the final ~~step~~ measurement step. Such sample processing operations like reagent addition, dilution, incubation, mixing, dialysis, extraction and many others can be implemented upto the point of sample introduction and detection.

Laboratory Robots

Laboratory robots are used to process samples when it might be dangerous for humans to be involved or when a larger number of routine steps might be involved. The robotic system is controlled by a computer programmed by the user. Laboratory robots can be used to dilute, filter, partition, grind, centrifuge, homogenize, extract, and treat samples with reagents. They can also be trained to heat and shake samples, dispense measured volumes of liquid, inject samples into chromatographic columns, weigh samples, and transport them to the appropriate instrument for measurement.

Advantages of Automated Systems

- It can lead to more analyses per unit time
- It helps when a large number of routine steps are involved
- They are used when it might be dangerous for humans to be involved.
- It has higher reliability, accuracy is highly obtainable
- Lower cost than manual when large samples are involved.
- ~~analytical~~ Disadvantages errors are limited

- without careful calibration it may lead to error
- High cost of maintenance
- once the analyses had started it cannot be checked or stop when ~~can~~ ^{cause} contamination when wrong sample ~~was~~ ^{'s} used. or even may affect the whole entire process
- they are very expensive